Self-Check Questions: Implicit Differentiation and Differential Equations

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1. What does the equation
$$F(x, y) = 0$$
represent geometrically?

   Answer. It represents a *curve* (or more precisely, a level set) in the xy-plane: the set of all points (x, y) for which F vanishes. For a "nice" function F this is typically a one-dimensional curve, but it can consist of several disconnected pieces, cross itself, or reduce to a single point.

2. How does an implicit curve differ from an explicit function
$$y = f(x)?$$

   Answer. An explicit function assigns exactly one y-value to each x; its graph passes the vertical-line test. An implicit curve $F(x, y) = 0$ imposes only a *relation* between x and y, so a single value of x may correspond to several values of y (e.g. both $y = 4$ and $y = -4$ lie on the circle $x^2 + y^2 = 25$ when $x = 3$).

3. Why can a single implicit equation $F(x, y) = 0$ represent multiple branches of a curve?

   Answer. One equation in two unknowns constrains the solution set to a one-dimensional object, but does not force it to be a graph. For a given x_0 the equation $F(x_0, y) = 0$ can have multiple roots, each root corresponding to a different branch of the curve stacked above x_0.

4. What assumptions are required for implicit differentiation to be valid?

   Answer. Near the point of interest one needs: (i) F has continuous partial derivatives F_x and F_y; (ii) $F_y \neq 0$ at the point (so the Implicit Function Theorem guarantees a local smooth branch $y = f(x)$); (iii) consequently, y is a differentiable function of x in a neighbourhood of that point.

5. How do you compute
$$\frac{dy}{dx}$$
from an equation such as
$$x^2 + y^2 = 25?$$

   Answer. Differentiate both sides with respect to x, treating y as a function of x:
$$2x + 2y\,\frac{dy}{dx} = 0.$$

Solving for dy/dx gives

$$\frac{dy}{dx} = -\frac{x}{y}.$$

6. Express $\frac{dy}{dx}$ in terms of the partial derivatives F_x and F_y for a curve defined by $F(x, y) = 0$, and derive the formula by differentiating both sides with respect to x.

   Answer. Differentiating the identity $F(x, y(x)) = 0$ with respect to x via the chain rule:

   $$\frac{\partial F}{\partial x} + \frac{\partial F}{\partial y}\frac{dy}{dx} = 0.$$

   Solving (assuming $F_y \neq 0$):

   $$\frac{dy}{dx} = -\frac{F_x}{F_y}.$$

7. Why does differentiating y^2 produce $2y\,y'$ rather than just $2y$?

   Answer. Because y is itself a function of x. The chain rule requires multiplying by the derivative of the inner function:

   $$\frac{d}{dx}\left[y(x)^2\right] = 2y(x) \cdot y'(x).$$

   The result $2y$ without y' would only be correct if y were an *independent* variable.

8. What is the geometric interpretation of $\frac{dy}{dx}$ computed by implicit differentiation, and how does it relate to the tangent line to the curve $F(x, y) = 0$ at a point?

   Answer. dy/dx evaluated at (x_0, y_0) is the *slope of the tangent line* to the curve at that point. The tangent line is

   $$y - y_0 = \left.\frac{dy}{dx}\right|_{(x_0,y_0)} (x - x_0).$$

   It is the best linear approximation to the curve near (x_0, y_0).

9. At which points does the formula for the implicit derivative fail to exist?

   Answer. Wherever $F_y = 0$, the formula $dy/dx = -F_x/F_y$ is undefined. Such points are *singular points* of the curve; the tangent line may be vertical, the curve may cross itself, or the IFT may break down entirely.

10. How can you determine whether a curve has a vertical tangent using implicit differentiation?

    Answer. A vertical tangent occurs where $dx/dy = 0$, i.e. where $-F_y/F_x = 0$, i.e. where $F_y = 0$ but $F_x \neq 0$. Equivalently, dy/dx is undefined ($F_y = 0$) but the curve is smooth (a regular point in the x-direction).

11. How do you compute $\frac{d^2y}{dx^2}$ for a curve defined implicitly by $F(x, y) = 0$? Illustrate the procedure using $x^2 + y^2 = r^2$.

    Answer. Differentiate the expression for y' once more with respect to x, substituting y' wherever it reappears.

For $x^2 + y^2 = r^2$ we found $y' = -x/y$. Differentiating:

$$y'' = \frac{d}{dx}\left(-\frac{x}{y}\right) = -\frac{y - xy'}{y^2} = -\frac{y - x(-x/y)}{y^2} = -\frac{y^2 + x^2}{y^3} = -\frac{r^2}{y^3}.$$

12. Given a one-parameter family of curves

$$F(x, y, c) = 0,$$

why does differentiating often eliminate the parameter c?

Answer. Differentiating with respect to x treats c as a constant, so its derivative is zero. The differentiated equation involves x, y, y', and possibly c. One can then use the original equation $F = 0$ to solve for c and substitute, or—if c already drops out after differentiation—the parameter is gone directly.

13. Why does eliminating one arbitrary constant usually produce a first-order differential equation?

Answer. One differentiation introduces y' (but no higher derivatives) and yields one additional equation. Using the original equation to eliminate the single constant c leaves a relation involving only x, y, and y'—precisely a first-order ODE.

14. Starting from

$$x^2 + y^2 = c^2,$$

derive the corresponding differential equation.

Answer. Differentiate with respect to x:

$$2x + 2y\,y' = 0 \implies y' = -\frac{x}{y}.$$

The constant c has disappeared; the ODE is $y' = -x/y$.

15. Starting from

$$y = ce^x,$$

derive the corresponding differential equation.

Answer. Differentiate:

$$y' = ce^x.$$

But $ce^x = y$, so the ODE is $y' = y$.

16. Starting from

$$y = x^2 + c,$$

derive the corresponding differential equation.

Answer. Differentiate:

$$y' = 2x.$$

The constant vanishes immediately; the ODE is $y' = 2x$.

17. Given a family of curves depending on n arbitrary constants, what systematic procedure eliminates all constants to produce a differential equation of order n?

    Answer. Write the family as $F(x, y, c_1, \ldots, c_n) = 0$. Differentiate successively n times with respect to x, obtaining equations $F^{(1)} = 0, \ldots, F^{(n)} = 0$ involving $y', y'', \ldots, y^{(n)}$ alongside the constants. The $n+1$ equations $F = 0, F^{(1)} = 0, \ldots, F^{(n)} = 0$ contain n unknowns $c_1, \ldots, c_n$; eliminate them (by substitution or resultants) to get a single ODE of order n in x, y, and $y', \ldots, y^{(n)}$.

18. Why do arbitrary constants disappear when differentiating?

    Answer. The derivative of a constant is zero. An additive constant simply vanishes; a multiplicative constant can often be expressed in terms of y (or a lower derivative) using the original equation, allowing it to be substituted out.

19. Is it possible for different curve families to generate the same differential equation?

    Answer. Yes. Different families may share the same ODE if they are all solution sets of that ODE. A first-order ODE typically has a one-parameter general solution, but it may also possess *singular solutions* (e.g. envelopes) that do not belong to that family yet still satisfy the equation pointwise.

20. What information is lost when constants are eliminated from a solution family?

    Answer. The specific member of the family—i.e. the particular solution. The constant encodes which initial condition was imposed, or equivalently which curve in the family the solution lies on. The ODE retains only the *shape* (direction field) shared by all members, not the identity of any individual curve.

21. What is meant by a solution of a differential equation?

    Answer. A function $y = \varphi(x)$ defined on some interval I, differentiable as many times as the order of the ODE requires, such that substituting φ and its derivatives into the equation produces an identity on I.

22. What is the difference between a particular solution and a general solution?

    Answer. The *general solution* contains all n arbitrary constants (for an nth-order ODE) and represents the entire family of solutions. A *particular solution* assigns specific numerical values to those constants, selecting one member of the family—typically the one satisfying given initial or boundary conditions.

23. Why does solving a first-order differential equation typically introduce one arbitrary constant?

    Answer. Solving requires one integration, which introduces one constant of integration. Equivalently, a first-order ODE describes slopes; specifying a slope field determines the shape of solution curves up to a vertical shift (or other one-parameter adjustment), which is fixed by one initial condition.

24. Why does solving a second-order differential equation typically introduce two arbitrary constants?

Answer. Two integrations are needed to pass from y'' back to y, each introducing one constant. Geometrically, a second-order ODE prescribes curvature; knowing both position and velocity at a point (two conditions) is required to pin down a unique solution.

25. What is the relationship between the order of a differential equation and the number of constants in its general solution?

    Answer. For a well-behaved ODE, the general solution of an nth-order equation contains exactly n arbitrary constants. This reflects the n-fold integration needed to reduce the equation to y, each step contributing one constant.

26. What is a separable differential equation? Given

$$\frac{dy}{dx} = \frac{g(x)}{h(y)},$$

    how do you find the general solution by separating variables?

    Answer. A separable ODE is one that can be written so that all y-dependence is on one side and all x-dependence on the other. For $dy/dx = g(x)/h(y)$, rewrite as

$$h(y)\,dy = g(x)\,dx,$$

    then integrate both sides:

$$\int h(y)\,dy = \int g(x)\,dx + C.$$

    The result (after solving for y if possible) is the general solution.

27. Given

$$y' = -\frac{x}{y},$$

    recover the family of solution curves.

    Answer. Separate variables: $y\,dy = -x\,dx$. Integrate:

$$\frac{y^2}{2} = -\frac{x^2}{2} + C_1 \implies x^2 + y^2 = C \quad (C = 2C_1 > 0).$$

    The solution family is the set of circles centred at the origin.

28. How can you verify that a given family of curves satisfies a differential equation?

    Answer. Substitute $y = f(x, c)$ (and its required derivatives, computed explicitly) into the ODE and check that the equation holds as an *identity* in x for every admissible value of c.

29. What is meant by an integral curve of a differential equation?

    Answer. An *integral curve* is a single solution curve of the ODE drawn in the xy-plane—a curve whose tangent direction at every point (x, y) agrees with the slope $f(x, y)$ prescribed by the equation. Each particular solution corresponds to one integral curve.

30. What is a slope field, and how does it represent the value of y' at each point in the plane?

    Answer. A slope field (or direction field) is a plot over a grid of points (x_i, y_j) of short line segments, each with slope $y' = f(x_i, y_j)$. It gives a visual picture of all directions that integral curves must have; solution curves are precisely those that are everywhere tangent to these segments.

31. Why do all curves in a solution family share the same tangent directions at corresponding points?

Answer. Because they all satisfy the same ODE $y' = f(x, y)$. At any point (x_0, y_0), the ODE dictates a unique slope $f(x_0, y_0)$, regardless of which member of the family passes through that point. Hence every integral curve through (x_0, y_0) must have the same tangent there.

32. For the equation

$$y' = -\frac{x}{y},$$

what does the slope field look like near the coordinate axes?

Answer.

- Near the y-axis ($x \approx 0$): slopes are close to 0; segments are nearly horizontal.
- Near the x-axis ($y \approx 0$): $|-x/y| \to \infty$; segments become nearly vertical (the formula is undefined on the x-axis itself).

This is consistent with the integral curves being circles: tangent lines are horizontal at the top/bottom of each circle and vertical at its leftmost/rightmost points.

33. In what sense does a differential equation carry the geometric content of an entire curve family, even though it contains no free parameter c?

Answer. The ODE encodes the slope at *every* point of the plane; any curve that everywhere matches these slopes is an integral curve. Thus the ODE implicitly encodes all members of the family simultaneously through the direction field—even without naming the constant c, it determines which directions are allowed, and the family is simply the collection of all curves that obey these directions.

34. Why can a differential equation be viewed as a rule assigning a direction to each point in the plane?

Answer. Writing $y' = f(x, y)$, the right-hand side assigns to each point (x, y) the number $f(x, y)$, which specifies the slope—equivalently, the direction $(1, f(x, y))$ up to scaling—of any solution curve passing through that point. This is exactly a *vector field* (or direction field) on the plane.

35. What is the statement of the Implicit Function Theorem?

Answer. Let $F : \mathbb{R}^2 \to \mathbb{R}$ be continuously differentiable near (x_0, y_0) with $F(x_0, y_0) = 0$ and $\frac{\partial F}{\partial y}(x_0, y_0) \neq 0$. Then there exist an open interval $U \ni x_0$ and a unique continuously differentiable function $f : U \to \mathbb{R}$ such that $f(x_0) = y_0$ and

$$F(x, f(x)) = 0 \quad \text{for all} x \in U.$$

Moreover, $f'(x) = -F_x(x, f(x))/F_y(x, f(x))$.

36. How does the Implicit Function Theorem justify implicit differentiation?

Answer. The IFT guarantees that, near a point where $F_y \neq 0$, the relation $F(x, y) = 0$ does define y as a *differentiable* function of x. It is therefore legitimate to differentiate the identity $F(x, y(x)) = 0$ term-by-term with respect to x using the chain rule, yielding $F_x + F_y y' = 0$ and hence $y' = -F_x/F_y$.

37. Under what conditions can an implicit relation $F(x, y) = 0$ be locally expressed as

$$y = f(x)?$$

Answer. It suffices that F is continuously differentiable near the point (x_0, y_0), that $F(x_0, y_0) = 0$, and that $\frac{\partial F}{\partial y}(x_0, y_0) \neq 0$. These are exactly the hypotheses of the Implicit Function Theorem.

38. Why can an implicit curve fail to be the graph of a function globally but still be a function locally?

Answer. Globally, the zero set of F may wrap around, have multiple branches over the same x-values, or self-intersect—all of which violate the vertical-line test. But at any point where $F_y \neq 0$, the IFT cuts out a small neighbourhood in which only one branch passes, and on that patch the curve *is* the graph of a smooth function.

39. What happens at points where

$$\frac{\partial F}{\partial y} = 0?$$

Answer. The IFT hypothesis fails; we can no longer guarantee a local smooth branch $y = f(x)$ near such a point. Geometrically, these are *singular points*: the tangent line may be vertical (if $F_x \neq 0$), or the curve may cross itself, develop a cusp, or be isolated. The implicit-derivative formula $-F_x/F_y$ is undefined there.

40. Given a two-parameter family

$$F(x, y, c_1, c_2) = 0,$$

how many times must you differentiate to eliminate both constants, and what is the order of the resulting differential equation?

Answer. You must differentiate *twice*, producing equations $F^{(1)} = 0$ and $F^{(2)} = 0$ that introduce y' and y''. Together with the original equation $F = 0$ you have three equations for the two unknowns c_1, c_2; eliminating them yields a second-order ODE.

41. Starting from

$$y = c_1 e^x + c_2 e^{-x},$$

derive the differential equation satisfied by the family.

Answer. Differentiate twice:

$$y' = c_1 e^x - c_2 e^{-x}, \qquad y'' = c_1 e^x + c_2 e^{-x} = y.$$

Hence $y'' = y$, or $y'' - y = 0$.

42. Can every differential equation be obtained by eliminating constants from an explicit solution family?

Answer. No. Some ODEs possess *singular solutions* (e.g. the envelope of the family) that satisfy the equation but cannot be obtained from the general solution by any choice of constants. Furthermore, many ODEs have no closed-form solution family at all; they are still well-defined equations, not derivable from explicit families.

43. What does it mean for a differential equation to define a family of curves rather than a single curve?

Answer. A first-order ODE specifies the slope y' at each point but does not fix which curve passes through a given point. Each choice of integration constant c selects a different integral curve; collectively these integral curves—one for each c—form a one-parameter family that "fills" the plane (under good conditions without crossings), and the ODE governs all of them simultaneously.

44. How do initial conditions select one member of a solution family?

Answer. An initial condition $y(x_0) = y_0$ requires the solution curve to pass through the specific point (x_0, y_0). Substituting into the general solution $y = f(x, c)$ gives one equation for c, which (under Picard–Lindelöf conditions) has a unique solution, singling out one member of the family.

45. Why is

$$y(0) = 1$$

sufficient to determine a unique solution of many first-order ODEs?

Answer. The general solution of a first-order ODE contains exactly one arbitrary constant c. Imposing $y(0) = 1$ yields a single equation $f(0, c) = 1$ for c. Provided f is smooth in c and the equation has a unique root (which the Picard–Lindelöf theorem guarantees locally), c is uniquely determined, pinning down a unique solution.

46. State the Picard–Lindelöf theorem. What conditions on f ensure that the initial value problem $y' = f(x, y)$, $y(x_0) = y_0$ has a unique solution on some interval?

Answer. **Picard–Lindelöf theorem.** Suppose $f(x, y)$ is continuous on a rectangle $R = \{|x - x_0| \le a,\ |y - y_0| \le b\}$ and satisfies a *Lipschitz condition* in y: there exists $L > 0$ such that

$$|f(x, y_1) - f(x, y_2)| \le L|y_1 - y_2| \quad \text{for all } (x, y_1), (x, y_2) \in R.$$

Then there exists $h > 0$ such that the IVP $y' = f(x, y)$, $y(x_0) = y_0$ has a *unique* solution on $|x - x_0| \le h$. (Continuity of $\partial f/\partial y$ on R is a sufficient condition for the Lipschitz hypothesis.)

47. Why are two conditions typically needed for a second-order ODE?

Answer. The general solution contains two arbitrary constants. Two independent conditions (e.g. $y(x_0) = y_0$ and $y'(x_0) = v_0$) supply two equations that uniquely determine both constants, selecting a unique solution. Specifying only one condition leaves one constant free, yielding infinitely many solutions.

48. How do initial conditions and boundary conditions both serve to determine integration constants, and what distinguishes the two?

Answer. Both impose constraints that fix the free constants in the general solution. The distinction is where the conditions are applied:

- *Initial conditions* specify the values of y (and its derivatives up to order $n - 1$) all at the *same* point x_0. They match the physical setup of knowing the state of a system at one instant, and existence/uniqueness is guaranteed by Picard–Lindelöf.

- *Boundary conditions* specify values of y (or its derivatives) at *different* points—typically the two endpoints of an interval. They arise naturally in spatial problems; existence and uniqueness need not hold in general and require separate analysis (Sturm–Liouville theory, etc.).

49. Can you explain, without calculations, why implicit differentiation and solving differential equations are inverse processes?

    Answer. Starting from a family of curves $F(x, y, c) = 0$, *implicit differentiation* eliminates the constant c and condenses the entire family into a single ODE—moving from many curves to one equation. Conversely, *solving* the ODE reintroduces a constant of integration and expands the single equation back into a family of curves—moving from one equation to many curves. Differentiation destroys information (which specific curve) and produces the ODE; integration restores that information (up to an arbitrary constant) and produces the family. The two operations are thus inverse to each other in the sense that "differentiate-then-integrate" recovers the family (up to relabelling of c), and "integrate-then-differentiate" recovers the ODE.